Topology Meets Truth: Leveraging Persistent Homology in Deep Fake Identification

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Abstract

Deepfake images are becoming increasingly realistic, which makes it harder to trust what we see online and creates new challenges for digital security. In this project, I explore a different approach to deepfake detection by using topological data analysis to study the underlying structure of facial images. Instead of depending on neural networks that often memorize pixel patterns, this method examines how geometric and topological features emerge across filtration of an image. I applied four filtrations to grayscale images, computed persistent homology, and extracted features such as persistence landscapes, persistent entropy, and Wasserstein distance. These topological features were then used to train a Support Vector Machine classifier. The results show clear structural differences between real and AI generated images, including distinct clustering patterns in PCA space and consistent differences in persistence landscapes and Betti curves. The SVM achieved 80 percent accuracy and showed strong recall for detecting deepfakes. Overall, the findings suggest that topology based features offer a promising and interpretable direction for deepfake detection, capturing subtle patterns that are often missed by traditional pixel level models.

1 Introduction

The growing use of generative AI and its saturation across media platforms has raised concerns about digital privacy and trust. Deepfakes, which are digitally altered images or videos, present a serious security risk because they can be created, modified, and distributed by AI systems at a scale and speed that allow misinformation to reach millions of people in mere seconds. To counteract this, companies are actively developing deepfake detection algorithms to identify and reduce harmful content.

However, many of the current approaches are based heavily on data-driven LLMs and neural networks. While often effective, these models face limitations that hinder their generalization and scalability. Most neural networks and LLM-based detectors tend to memorize how a dataset looks by focusing on surfacelevel cues such as pixel behavior or noise patterns rather than learning the deeper structural patterns involved in generating synthetic images.

Topological data analysis (TDA) presents a robust solution to these limitations. This method constructs geometric structures from pixel data and extracts topological features that are then fed into a machine-learning model.

2 Data

In this project, I build on the framework from *Topological Data Analysis for Classification of AI Generated Faces* (2024). The model is trained on the Deepfake Face Image Classification Dataset by Paudel (Hugging Face), containing 16,060 AI-generated and 16,060 real images. In preprocessing, all images were resized to 256×256 and converted to grayscale.

3 Methods

To detect deepfakes, I implemented a multi-stage pipeline consisting of image preprocessing, topological feature extraction, and machine-learning classification. After preprocessing, each image is converted into a 256×256 grayscale pixel grid where each pixel takes a value between 0 and 1. This pixel grid forms the base space for applying filtrations.

3.1 Cubical Complex Construction

Since each image is represented as a pixel grid, cubical complexes provide a natural structure for building filtrations. Since pixels and the 2-cube complex structure have similar structures each pixel essentially joins the complex through various filtrations.

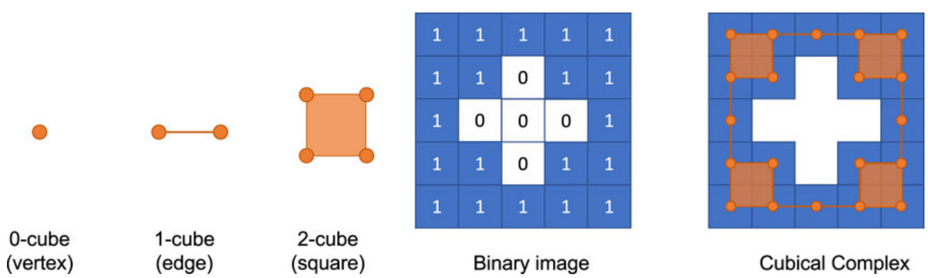

Figure 1: Cubical Complex

3.2 Filtrations

I designed a dedicated `Filtration` class and implemented four filtration functions that operate directly on the pixel space.

3.3 Height Filtration

The height filtration assigns each pixel a value equal to its grayscale intensity:

$$f_{\text{height}}(x) = I(x)$$

3.4 Radial Filtration

Radial filtration assigns each pixel a value equal to its Euclidean distance from the image center:

$$f_{\text{radial}}(x) = \|x - c\|_2$$

3.5 Density Filtration

Density filtration measures local grayscale intensity:

$$f_{\text{density}}(x) = \frac{1}{|N(x)|} \sum_{y \in N(x)} I(y)$$

3.6 Dilation Filtration

Dilation filtration assigns each pixel a value corresponding to the dilation step at which it becomes activated:

$$f_{\text{dilation}}(x) = \min\{r \mid x \in \text{Dilate}(A, r)\}$$

3.7 Persistent Homology

After constructing the filtrations via cubical complexes I analyze various topological features of the image. I first compute the homology groups H_0 (connected components) and H_1 (holes) for different filtration types.

3.8 Persistence Diagrams

By analyzing the homology across various successive filtrations on the pixel, I can then analyze the birth (b_i and death d_i of homology features across a filtration. The birth and death of these features can then be analyzed via `giotta-tda` library to generate image barcodes and persistence diagrams for each filtration. In the project I took a random subsample n = 40 images and produced 4 persistence diagrams and barcodes per image.

3.9 Persistence Landscapes

After generating barcodes and persistence diagrams, I then vectorized the diagrams into persistence landscapes to use as input features for my machine learning model. Each birth-death pair (b_i, d_i) is converted into a function:

$$\lambda_i(t) = \begin{cases} 0, & t \notin [b_i, d_i] \\ t - b_i, & b_i \le t \le \frac{b_i + d_i}{2} \\ d_i - t, & \frac{b_i + d_i}{2} \le t \le d_i \end{cases}$$

I computed the persistence landscape with giotta.tda and then sampled with .fit-transform into feature vectors for the model.

3.10 Persistent Entropy

Another feature I calculated was persistence entropy. This is a single scalar value that represents how spread apart homology features appear across the filtration. Persistence entropy is calculated by calculating the lifetime $p_i = d_i - b_i$, then normalized by $w_i = \frac{p_i}{\sum_j p_j}$, then persistent entropy is calculated by the following formula: $H = -\sum_i w_i \log w_i$.

I computed the persistence entropy for a sub sample n = 50 (25 real and 25 fake) images.

3.11 Wasserstein Distance

The last feature I calculated was the Wasserstein distance. The Wasserstein distance quantifies how far two persistence diagrams are from each other by computing the optimal matching between their birth–death points. In deepfake detection, this distance captures how strongly an image's topological structure deviates from the typical patterns found in real faces, making it a useful feature for distinguishing synthetic images.

I calculated the Wasserstein distance between between an empty and completed filtration persistence diagram.

3.12 Support Vector Machine

TDA features (landscapes, entropy, Wasserstein curves) are fed as input features into a Support Vector Machine (SVM) classifier. The SVM creates a linear boundary that separates the real and fake images based on the feature input space. I took a random subsample n = 200 and created a train and test training set. The training dataset a sample size n = 40 (25 real and 15 fake) and the test set contains a sample size of n = 160. I then calculated the precision, accuracy, recall, and F1 score.

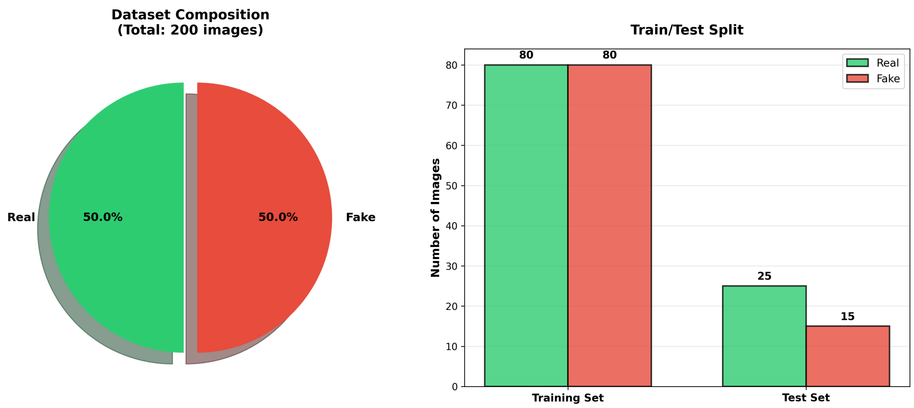

Figure 2: Enter Caption

4 Results

4.1 Filtrations

I generated filtration heatmaps for a subsample of 50 images consisting of 25 real and 25 fake examples. Each heatmap was normalized on a scale from 0 to 1 to represent the progression of the filtration from its earliest to latest stage. Using the four filtration methods, the heatmaps reveal distinct structural patterns in how different regions of an image enter the filtration. These visualizations help illustrate how intensity, spatial layout, local texture, and morphological growth differ between real and AI generated faces.

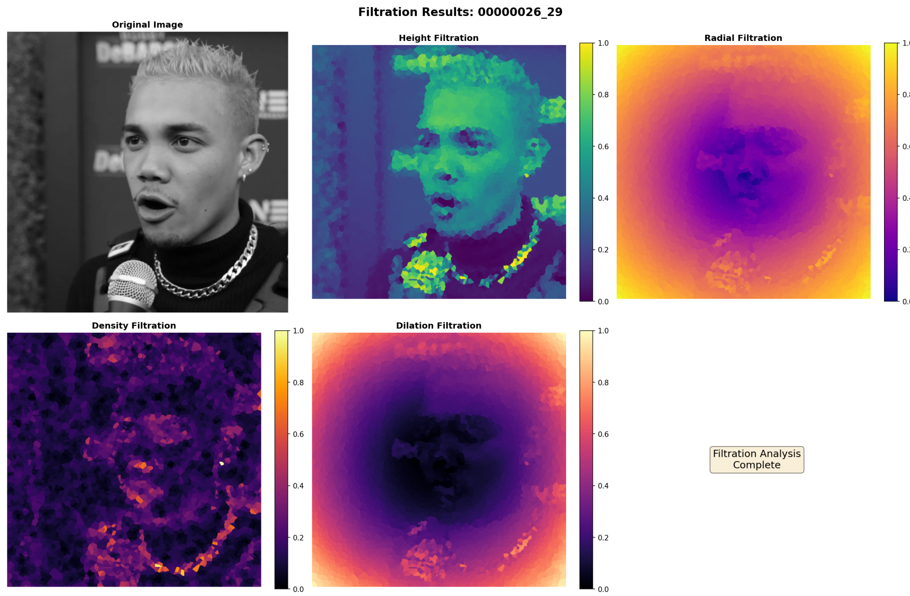

Figure 3: Filtration Results

4.2 Barcodes and Persistence Diagrams

There were 4 barcodes and 4 persistence diagrams calculated corresponding to the four filtration types leveraged in this project. From the analysis of the 40 sample images, real images tend to have more uniform births of H_0 features while deepfake generated images have more sporadic feature births. Additionally, both deepfake and authentic images tended to never produce any sufficient and long-lasting H_1 features. Either these features would never be generated or would birth and die extremely quickly which made them extraneous for analysis. The absense of the H_1 groups could be linked to the fact that pixels in images tend to not contain holes and cavities or as these H_1 regions they are filled in very quickly before they can become topological features of the filtration. Below are two images, deepfake and real, that show two sample H_0 persistence diagram and barcodes calculated for various filtration methods.

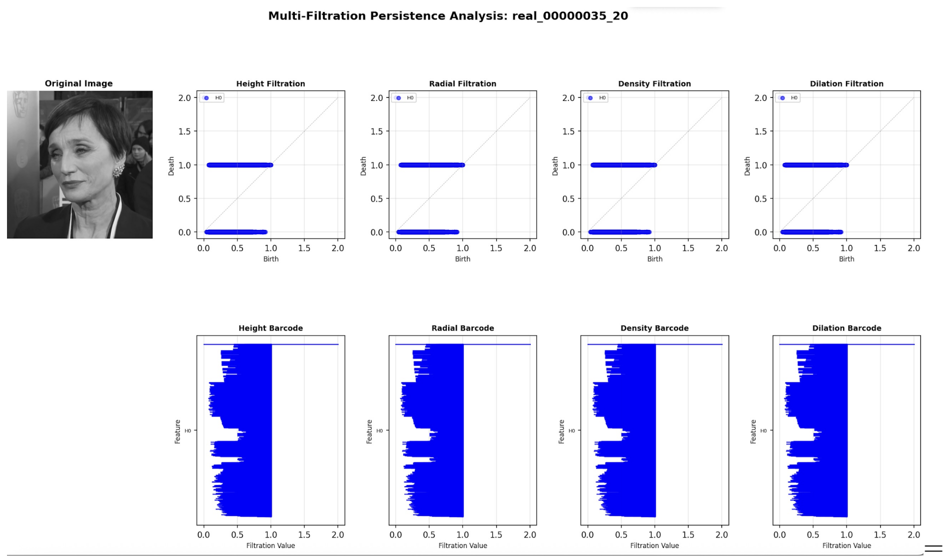

Figure 4: Authentic Image Persistence Diagrams and Barcodes

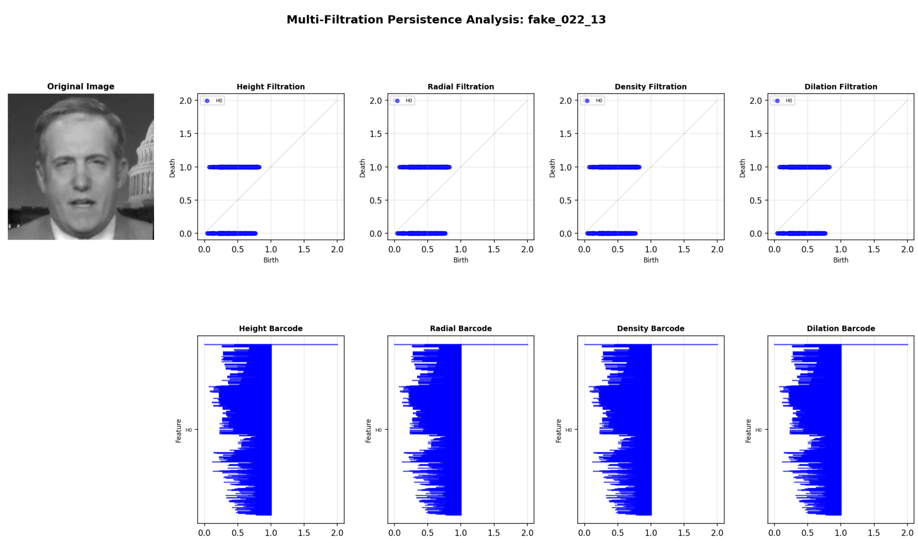

Figure 5: Deepfake Image Persistence Diagrams and Barcodes

4.3 Persistence Landscapes

The persistence landscapes of deepfake images tend to have peaks earlier than real images. A rational for this behavior is that deepfake algorithms tend to introduce smooth, globally consistent features that cause the H_0 features to peak quicker.

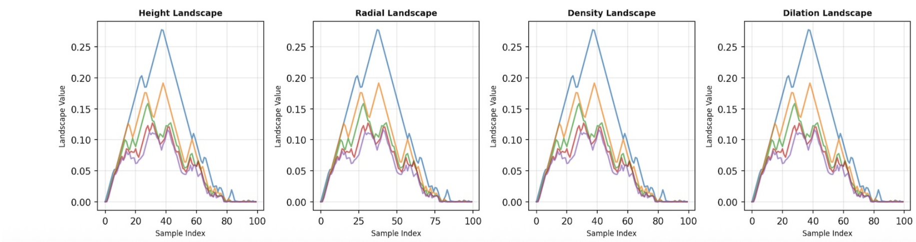

Figure 6: Real Image Persistence Landscapes

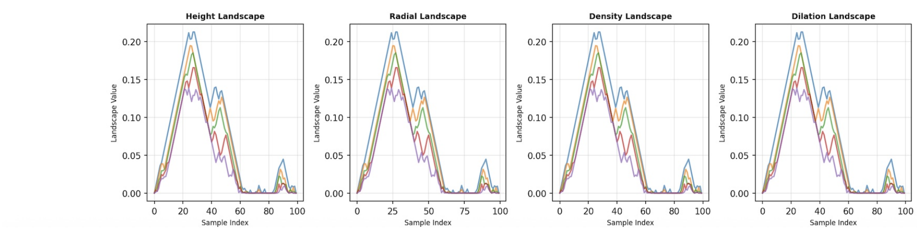

Figure 7: Deepfake Image Persistence Landscapes

4.4 Betti Curves

The Betti curves, particularly the H_1 and H_2 features, for the real images appear to decrease linearly. The H_1 and H_2 for the deepfake images appear to increase then decrease. These results regarding the H_1 and H_2 feature results are interesting because the persistence diagrams produced never appeared to generate H_1 and H_2 features. Since β_1, and β_2 measure the dimension of the homology groups at various filtration this could imply that features are extremely short-lived and that the birth and death may happen at only a moment in the filtration process. More research should be conducted to understand the behavior of the Betti curves, but these two distinct patterns β_1 and β_2 suggest Betti numbers would be optimal features for the model classifier.

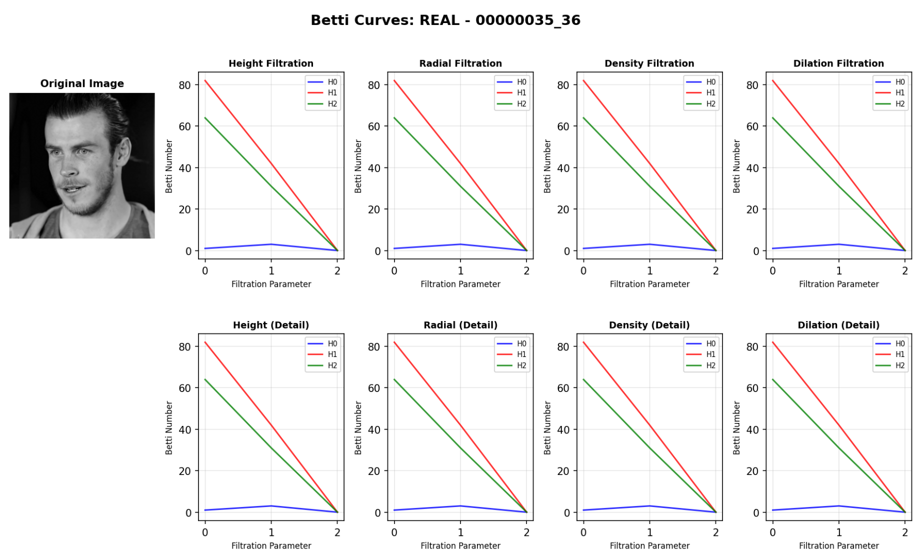

Figure 8: Real Image Betti Curves

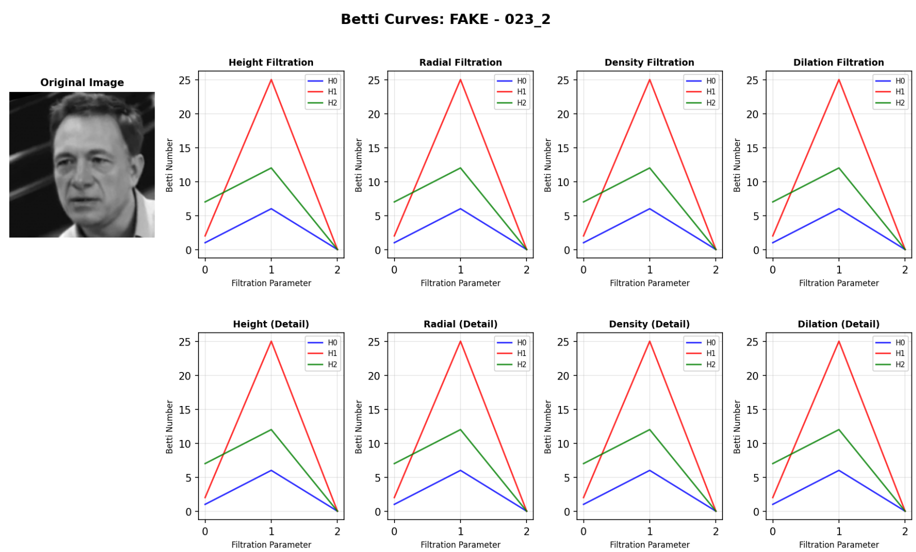

Figure 9: Deepfake Image Betti Curves

4.5 2D PCA of Persistence Landscape

To see if a TDA approach could be a suitable fit a deepfake detection model, I used PCA to analyze the dataset. I ran a 2D PCA to capture the variance and detect signals between deepfake and authentic images. I randomly sub sampled 600 images from the dataset (300 fake, 300 authentic) and applied PCA testing on persistence landscapes. The results are plotted in the figure below.

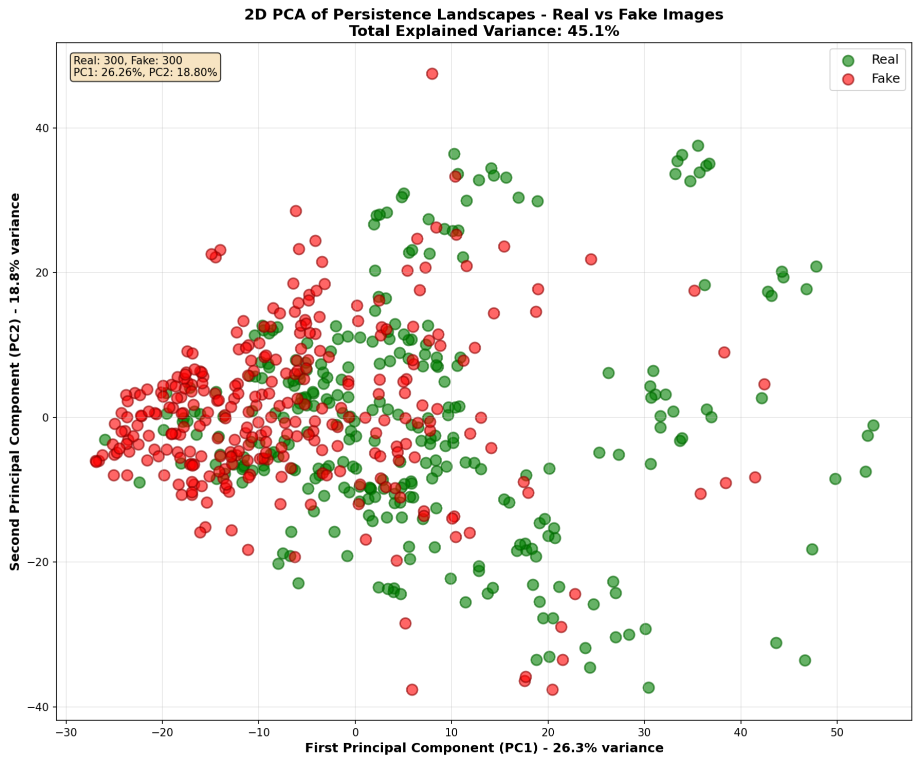

Figure 10: 2D PCA of Persistence Landscape

The results show signals of a clustering phenomenon that occurs with the fake images. The vast majority of the fake images appear clustered to the left, while the real images tend to show a wider spread. These results suggest that deep fake images tend to have similar shared topological features while real images tend to have larger spread and variance in there underlying topology. The clustering phenomena seen in the fake images suggest that a TDA feature extraction approach can be suitable for detecting fake images.

4.6 SVM Results

The SVM model performed reasonably well on the test set of 40 images. According to the confusion matrix, it correctly classified 13 fake images and 19 real images, resulting in an overall accuracy of 80%. The model was particularly effective at detecting deepfakes, achieving a recall of 87%, which indicates that it successfully identified most of the fake images in the evaluation set.

The precision for the real class was also strong at 90.48%, meaning that predictions labeled as real were accurate the majority of the time. The primary challenge emerged in differentiating real images from fakes, as reflected by the six false negatives. This misclassifications pattern may occur when real images contain smoother or more uniform texture patterns that resemble deepfake-like structural characteristics after undergoing TDA-based feature extraction.

Overall, these results suggest that topological features such as persistence landscapes, persistent entropy, and Wasserstein distance provide meaningful structural information for distinguishing real and synthetic faces. While the model is not flawless, its performance demonstrates that TDA-derived features capture subtle geometric and topological differences between authentic and AI-generated images, supporting the viability of this approach for deepfake detection.

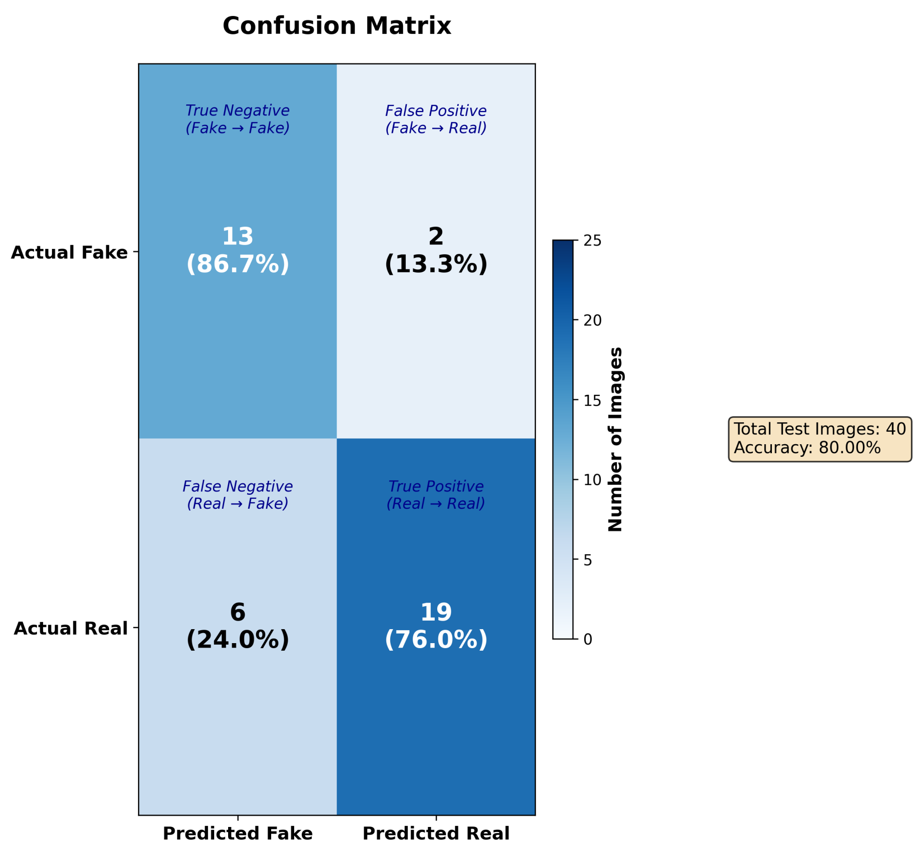

Figure 11: Confusion Matrix

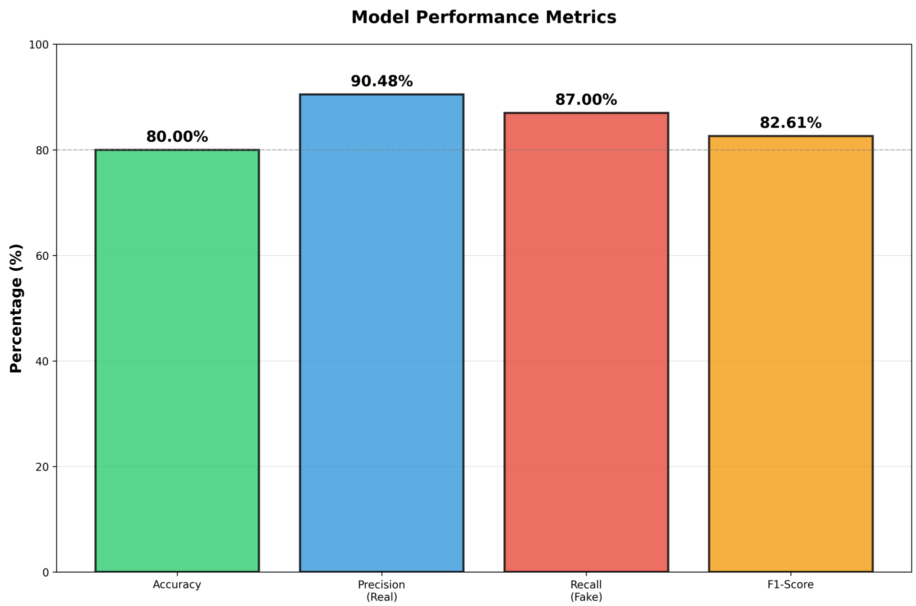

Figure 12: Model Metrics

5 Discussions

The results of this project show that topological data analysis can capture meaningful structural differences between real and AI generated facial images. Across filtrations, the persistence landscapes and PCA plots revealed that deepfake images tend to share more uniform and tightly clustered topological patterns, while real images display much greater variation. This suggests that generative models often introduce smooth and globally consistent structures that TDA can detect. The Betti curves also reflected this pattern. Deepfake images showed brief and fluctuating higher dimensional features, while real images followed a more predictable and stable decline across the filtration.

The classification results support the value of this approach. Even with a relatively small training sample, the SVM achieved an accuracy of 80 percent and showed strong recall for detecting deepfakes. Most errors came from real images being marked as fake, which may happen when natural images contain smooth regions that resemble synthetic structure after filtration. The performance indicates that persistence landscapaes, entropy, and Wasserstein distance capture enough global information to separate real and fake images without depending on pixel level artifacts.

Overall, these findings suggest that TDA provides a promising and interpretable framework for deepfake detection. Instead of focusing on local texture cues, TDA highlights how the global organization of an image evolves throughout the filtration process. This allows it to detect subtle structural differences between authentic and AI generated faces. While future work should expand the dataset and explore additional filtrations, the results here show that topology based features have real potential for building reliable and transparent deepfake detection systems.

6 Sources

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7 AI Acknowledgment

I would like to acknowledge the use of ChatGPT and Cursor as AI tools throughout the development of this project. Cursor was used to assist with programming tasks, including debugging, structuring Python files, and organizing the codebase for my filtration pipeline and feature extraction modules. ChatGPT provided support in the planning stages of the project by helping me outline my research approach, clarify topological concepts, and refine the explanations of methods. I also used ChatGPT to help polish written sections, improve clarity, and format my LaTeX document on Overleaf. All analytical decisions, coding logic, and interpretations of results were carried out by me, with these tools serving as guidance and writing support during the process.